

Anton Melnychuk

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EDUCATION

Yale University

B.S. in Electrical Engineering & Computer Science (GPA 3.8/4)
M.S. in Computer Science (Sterling Prize Fellowship)

New Haven, CT
May 2026
Expected Jan 2028

SUMMARY

Most excited about hardware-software co-design for large-scale compute infrastructure. Hands-on experience in architecture, networking/interconnects, hardware acceleration, verification, and tapeout flows.

Languages: SystemVerilog, C, Rust, Python, Tcl.
Tools: Xilinx FPGAs (Zynq/Versal), Cadence, Vivado/Vitis, KiCad.

WORK EXPERIENCE

Riverlane Ltd.

Quantum Research Intern

Greater Boston, MA
May 2026 – Present

- [Deltaflow v3](#): Next-Generation Quantum Error Correction Infra.
 - Designed and hardware-accelerated distributed graph decoders for topological and surface-codes.
 - FPGA-porting NVIDIA Ising, achieving 1.5x logical-error-rate improvement at sub-1 μ s latency.

Skills: graph/pipeline decoding, low-latency systems, latency profiling, memory placement, GPU/HPC, distributed systems, timing closure & STA, high-speed SerDes & signal integrity, RTL/FPGA design (SystemVerilog, Vivado).

Microsoft Research Asia

Research Collaborator (External)

Canada, Remote
Mar. 2026 – Present

- [FengHuang](#): Next-Generation Memory Orchestration for AI Inferencing.
 - LLM workloads outgrow HBM storage; first to solve by decoupling memory from compute.
 - RTL designed Tensor Addressable Bridge (TAB), prefetch silicon interfacing with NVIDIA B200.

Skills: Advanced packaging (2.5D, CoWoS, UCIe), RTL design, SGLang, MoE, Cadence Genus, signal integrity, SerDes, high-bandwidth memory (HBM, LPDDR6), interconnects (NVLink, InfiniBand), AI inference systems.

Asynchronous VLSI Lab

Undergraduate Research (ASIC)

New Haven, CT
Jul. 2025 – May 2026

- [Neurae](#): iEEG responsive seizure neuromodulation brain-computer interface (ongoing patent).
 - Full RTL-to-GDSII Design and Verification of a DAQ, detection, and stimulation async NoC tiles.
 - Outperformed NeuroPace® patented technology (US6810285B2), reducing false positives.

Skills: Sync vs. Async RTL Design (NoC), SVA/UVM/TBs, DSP arrays, GSPS, CI/CD Synthesis, 2.5D packaging, Timing Closure/CDCs, ModelSim, EMIR/Power/Signal integrity analysis, Sign-offs, SoCs/NoC routing, RISC, Cadence, SPICE.

Efficient Computing Lab

Undergraduate Research (Xilinx FPGAs)

New Haven, CT
Sep. 2024 – Aug. 2025

AWARDS

Partcl x HRT Macro Placement Challenge (8/129 teams)
High School, Ukraine Global Scholar

2026
2021

SELECTED PROJECTS

Longhorn Silicon, UT Austin AI Inference Project

May 2026 – Present

- RTL-designed TurboQuant KV-cache coprocessor for [Lambda](#), targeting LLMs up to 200M parameters.

Wireless Mechanical Keyboard, Open-Source PCB Project

Jan. 2026 – Feb. 2026

- Designed and fabricated under sale an 80-key hotswap wireless [keyboard](#) PCB in KiCad.

Nintendo Entertainment System CPU, VLSI Final Project

Oct. 2025 – Dec. 2025

- Custom Physical Design of [MOS 6502 8-bit CPU](#) (powered Apple II & Nintendo Entertainment System).

PARCv2 Five-Stage Pipeline Processor, Computer Architecture Final Project

Mar. 2025 - Apr. 2025

- 33% speeding-up PARCv2 CPU with speculative fetch, dynamic scheduling, [reorder buffer](#) execution.

TEACHING

Teaching Assistant, CPSC 4200/ECE 4201 Computer Architecture, Yale University
Teaching Assistant, CPSC 3230 Systems Programming, Yale University

Spring 2026
Spring & Fall 2025